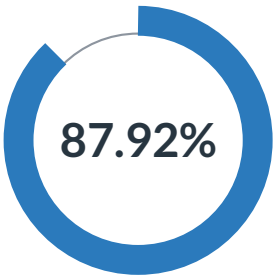
Results

Carter Feng



52.75
Out of 60 points

46:03
Time for this attempt

Your Answers:

1 0 / 0 points Fill in the Blank

At Seven Lakes High School, we believe that character matters and that academic integrity reflects who we are as learners and citizens. By signing this pledge, I commit to upholding the values of honesty, responsibility, and respect in all academic and personal endeavors.

My Academic Integrity Pledge:

- **I will complete all academic work honestly, without giving or receiving unauthorized assistance.**
- **I will not copy, plagiarize, or submit someone else's work as my own.**
- **I will not use unauthorized materials (notes, devices, online resources) during assessments.**
- **I will respect the integrity of others' work and will not aid others in academic dishonesty.**
- **I will not share or receive academic work, ideas, or answers electronically or in-person when individual work is expected or collaboration is not permitted.**
- **I will report any knowledge of cheating or violations of academic integrity to a trusted adult.**

I understand that academic dishonesty harms my character, undermines trust in our community, and may result in consequences outlined in the school's academic integrity policy.

Print your name to acknowledge the oath. Not printing your name will result in a zero for the assessment.

✓ Carter Feng

2 5 / 5 points Fill in the Blank

Create a Binary Search Tree.
Insert: 83 27 34 55 20 32 31 3 73 64 14 47 92 60 58
Delete: 83 55 3 14 34

How many nodes in the BST are of degree 0?

✓ 3

3 5 / 5 points Fill in the Blank

Create an AVL Tree.
Insert: 31 49 68 8 4 37 28 73 59 6 97 50 54 41 79
Delete: 54 73 31 6 8

What is the root?

✓ 49

4 1.25 / 5 points Matching

Create an AVL Tree.
Insert: 31 49 68 8 4 37 28 73 59 6 97 50 54 41 79
Delete: 54 73 31 6 8

What are the following nodes values when the 5th rotation occurs?

What type of rotation is it?	✓	left - right
Ancestor Node	✗	97 Correct Answer: **59**
Child Node	✗	49 Correct Answer: **54**
Parent Node	✗	28 Correct Answer: **50**

5 5 / 5 points Matching

```
Create an AVL Tree.
Insert: 31 49 68 8 4 37 28 73 59 6 97 50 54 41 79
Delete: 54 73 31 6 8

What are the following nodes values when the 6th rotation occurs?
```

Child Node	✓	49
What type of rotation is it?	✓	right - left
Parent Node	✓	68
Ancestor Node	✓	37

6 5 / 5 points Fill in the Blank

Create an AVL Tree.
Insert: 31 49 68 8 4 37 28 73 59 6 97 50 54 41 79
Delete: 54 73 31 6 8

How many nodes in the AVL tree have a balance factor of −1?

✓ 1

7 5 / 5 points Essay

Write the rightRightRotation method for an AVLTree class.

```
private AVLNode rightRightRotation(AVLNode a, AVLNode p)
{
```

```
public AVLNode rightRightRotation(AVLNode a, AVLNode p) {
    AVLNode right = a.right();
    a.setRight(right.left());
    right.setLeft(a);
    return right;
}
```

Correct

8 2.5 / 5 points Matching

Create a red-black tree.
Insert: 59 79 72 92 13 83
Delete: 59
Insert: 60 47 14 64
Delete: 79 60
Insert: 87 34 44 46 6
Delete: 13 47

What is the preorder traversal of all the red nodes in the final tr
Enter null when nothing is present.
Only enter value, not color.
Enter values in row major order.

1 2 3 4 5 6 7 8 9 10

8	✓ null	
6	✓ null	
4	✗ 87	Correct Answer: **null**
5	✗ 83	Correct Answer: **null**
10	✓ null	
2	✗ 34	Correct Answer: **46**
3	✗ 46	Correct Answer: **83**
7	✓ null	
1	✗ 14	Correct Answer: **34**

9

✓ null

9 5 / 5 points Essay

Write the rightRightRotation method for a RedBlackTree class.

```
private RedBlackNode rightRightRotation(RedBlackNode g, RedBlackNod
{
```

```
    private RedBlackNode rightRightRotation(RedBlackNode g, RedBlackNode p) {
        g.setRight(p.left());
        p.setLeft(g);
        g.setColor(g.getColor() - 1);
        p.setColor(p.getColor() + 1);
        return p;
    }
```

Correct

10 5 / 5 points Matching

Create a red-black tree.
Insert: 59 79 72 92 13 83
Delete: 59
Insert: 60 47 14 64
Delete: 79 60
Insert: 87 34 44 46 6
Delete: 13 47

What are the following nodes values when the 1st color swap occurs?

Left Child	✓	59
Right Child	✓	79
Parent	✓	72

11 5 / 5 points Matching

```
Create a red-black tree.
Insert: 59 79 72 92 13 83
Delete: 59
Insert: 60 47 14 64
Delete: 79 60
Insert: 87 34 44 46 6
Delete: 13 47

What are the following nodes values when the 3rd color swap occurs?
```

Right Child	✓	34
Parent	✓	14
Left Child	✓	13

12 5 / 5 points Matching

```
Create a red-black tree.
Insert: 59 79 72 92 13 83
Delete: 59
Insert: 60 47 14 64
```

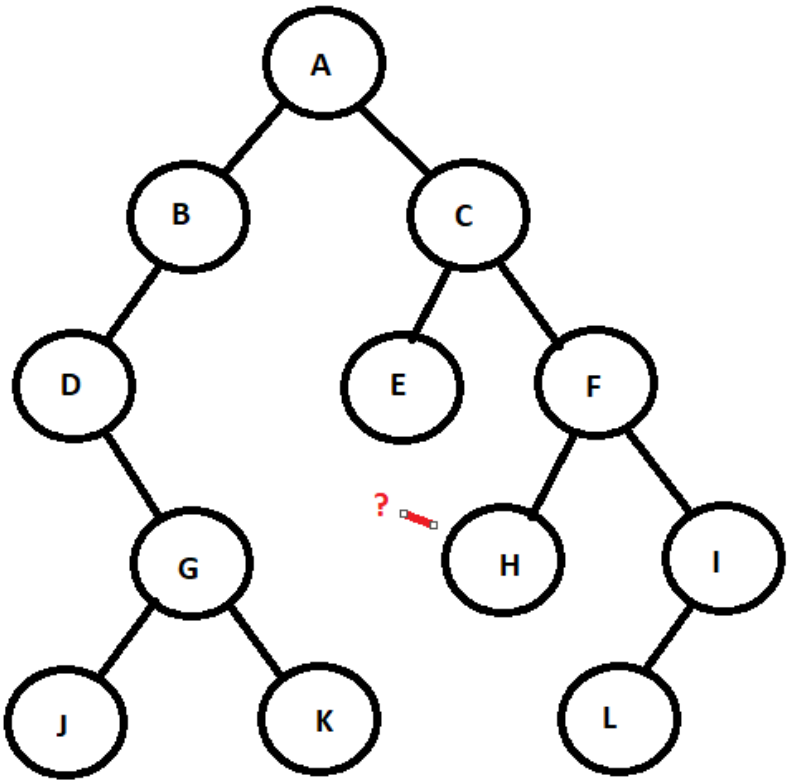
Delete: 79 60
Insert: 87 34 44 46 6
Delete: 13 47

What is on level 2 of the red-black tree?
Enter null when nothing is present.
Only enter value, not color.
Enter values in row major order.

1 2 3 4

2	✓ 44
1	✓ 6
4	✓ 92
3	✓ 72

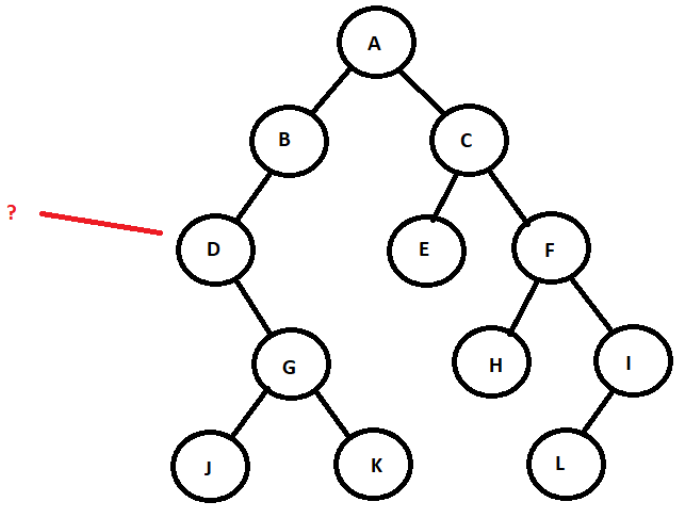
13 1 / 1 point Multiple Answer



Identify the marked item [?].

- [ ] parent, internal node, inner node, branch node
- [ ] root
- [ ] right child
- [x] left child
- [ ] path
- [ ] edge
- [x] leaf, outer node, terminal node
- [x] sibling

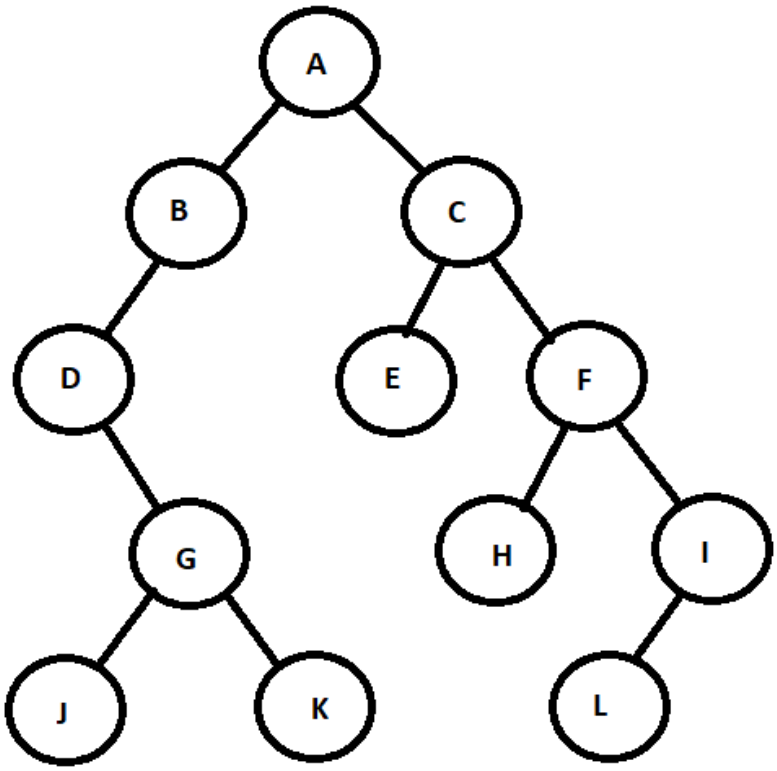
14 1 / 1 point Multiple Answer



Identify the marked item [?].

- [ ] right child
- [ ] edge
- [ ] path
- [ ] sibling
- [x] parent, internal node, inner node, branch node
- [x] left child
- [ ] leaf, outer node, terminal node
- [ ] root

15 1 / 1 point Fill in the Blank



What is the level location of node I

✓ 3

16 1 / 1 point Multiple Choice

What is the worst case big O time constraint for searching for a no

- O(log N)
- ✓ O(N)
- O(N^2)
- O(1)

○ `O(N log N)`

17 0 / 1 point Multiple Choice

What type of traversal prints the roots of each subtree from the same depths before printing the subtrees of the children?

○ postorder traversal

○ inorder traversal

× ◉ preorder traversal

Correct Answer: **level order traversal**

○ level order traversal